Project: Worked examples
Title: Insulation thickness (ISO 12241 formula route)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

The insulation thickness on a pipe (cylinder) or a flat surface that meets a heat-flow limit, a limit on the surface temperature of a hot surface and an anti-condensation requirement for a cold one, each solved separately and the governing one reported; and for a chosen thickness the heat flow, the surface temperature and PASS/FAIL per criterion. Steady state, one layer, no thickness table of any standard.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Pipe resistance per metre | $R = \ln(D_e/D_i)/(2\pi\cdot\lambda) + 1/(h_{se}\cdot\pi\cdot D_e)$, $D_e = D_i + 2t$; with the inner film and pipe wall $R_{oth} = 1/(h_i\cdot\pi\cdot D_p) + \ln(D_i/D_p)/(2\pi\cdot\lambda_w)$ added in series |
| Flat resistance per square metre | $R = t/\lambda + 1/h_{se}$ (+ $1/h_i + t_w/\lambda_w$) |
| Heat flow | $q = (\theta_i - \theta_a)/R$, positive from the process to the ambient |
| Surface temperature | $\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ (pipe), $\theta_a + q/h_{se}$ (flat) |
| Surface coefficient | $h_{se}$ as given, or $h_{se} = h_c + h_r$ with $h_r = \varepsilon\cdot\sigma\cdot(T_s^2 + T_a^2)(T_s + T_a)$; $h_c$ from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with $\theta_{se}$, which it depends on |
| Heat-loss limit | the thickness beyond which $\lvert q\rvert \le q_{max}$; surface-temperature limit (hot): $\theta_{se} \le \theta_{se,max}$; anti-condensation (cold): $\theta_{se} \ge \theta_{dew}$ + margin, $\theta_{dew}$ entered (for example from B2) or from the Magnus relation at $\theta_a$ and RH |
| Thickness solution | each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria |
| Critical diameter | $D_{crit} = 2\lambda/h_{se}$; below it, insulation raises the heat flow until $D_e$ passes $D_{crit}$ |

Basis: Thermal insulation for building equipment and industrial installations: calculation rules[1](formula route (clause numbers to be checked)); External forced, free and mixed convection correlations[2](7th ed., §7.2, §7.4, §9.6, §9.9 (to be checked)); Magnus saturation vapour pressure (shared with B7)[3](cl. 6 (clause number to be checked against the owner's copy)).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| geometry | Surface | pipe (cylinder) | | assumption: A round pipe. | any |
| $D_i$ | Pipe outer diameter, the inside of the insulation (pipe) | 90 | mm | drawing: M-PIP-014 rev A | 10 … 1000 mm |
| $\theta_i$ | Process (fluid) temperature | 6 | °C | spec: SPEC-M-11 §3: chilled-water flow temperature | -50 … 400 °C |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $\theta_a$ | Ambient air temperature (also the radiation sink) | 28 | °C | spec: SPEC-M-11 §2: plant-room design temperature | -20 … 50 °C |
| $\lambda$ | Thermal conductivity of the insulation | 0.035 | W/(m·K) | assumption: Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material) | 0.015 … 0.2 W/(m·K) |
| $h_{mode}$ | Where the outer surface coefficient comes from | h_se as given | | assumption: A surface coefficient is given, not computed. | any |
| $h_{se}$ | Outer surface coefficient $h_{se}$ (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there) | 8 | W/(m²·K) | assumption: Still air in a plant room; replace with the project's own record. | 3 … 40 W/(m²·K) |
| eps | Emissivity of the outer surface (for the correlation; with $h_{se}$ as given it only judges that value against condensation on a cold surface) | 0.9 | | assumption: Painted or plastic jacket (invented for this example); with h_se given it only judges that value against condensation. | 0.05 … 1 |
| $v_{wind}$ | Air speed over the surface (correlation; 0 = still air) | 0 | m/s | assumption: Not used: h_se is given. | 0 … 20 m/s |
| $L_{surf}$ | Height or flow length of the flat surface (correlation, flat) | 1 | m | assumption: Not used: a pipe. | 0.1 … 10 m |
| $q_{max,m}$ | Heat-loss limit per metre of pipe (generic default of 25 W/m, replace with your project limit; blank = not a criterion) | 12 | W/m | spec: SPEC-M-11 §5: heat gain not to exceed 12 W/m | 1 … 1000 W/m |
| $dew_{margin}$ | Margin of the surface above the dew point, for a cold surface (blank = not a criterion) | 1 | K | spec: SPEC-M-11 §5: surface at least 1 K above the dew point | 0 … 5 K |
| $RH_a$ | Relative humidity of the ambient air (anti-condensation) | 70 | % | spec: SPEC-M-11 §2: plant-room design humidity | 20 … 95 % |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $t_{chosen}$ | Chosen insulation thickness (blank = report at the required thickness, no checks; with it, nothing blocks) | 25 | mm | spec: Pipework schedule M-SCH-014: 25 mm sleeves | 0 ... 300 mm |

## 3. Assumptions

A1. Steady one-dimensional heat flow through a single homogeneous layer of constant conductivity; no thermal bridges at supports, no joints, no moisture in the insulation. λ is taken at the value entered: use the value at the mean insulation temperature shown on the sheet (optionally noted in 'Temperature the conductivity applies at'). (Formula route of the standard.)
A2. With the correlation, $h_{se} = h_c + h_r$. Radiation: $h_r = \varepsilon \cdot \sigma \cdot (T_s^2 + T_a^2)(T_s + T_a)$ to surroundings at the ambient air temperature. Free convection to air: Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms (horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3); vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3)). Forced convection: Churchill–Bernstein for a cylinder in cross-flow, the laminar/mixed flat-plate relations along a plate; air properties from fits at the film temperature, Pr = 0.71; blended with free convection as $(h_{nat}^3 + h_{forced}^3)^{(1/3)}$. Indicative, not the standard's own correlation. (Incropera 7th ed. §7.2, §7.4, §9.6 free convection and §9.9 mixed convection (section numbers to be checked against the owner's copy); the free-convection constants are Holman's, Heat Transfer; the standard's own coefficients are not reproduced.)
A3. The dew point is entered (for example from B2) or is the Magnus relation at $\theta_a$ and RH, the same closed form B7 uses. Below 0 °C the relation is over ice, so RH is then taken relative to ice, whereas meteorological RH is usually relative to water: enter the dew point for sub-zero air. A margin above it is the user's. (Derived; B7 shares the function.)
A4. Pipes are treated as horizontal for free convection and flat surfaces as vertical; air properties are fits (conductivity within about 3 %, viscosity within about 4 % over 250–400 K), Pr = 0.71. (Stated.)
A5. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m, which is what matters when thin insulation on a small pipe raises the heat flow (critical diameter). (Stated.)
A6. No thickness table is used or shipped: not BS 5422, not the tables of any standard. The conductivity is the user's input, from a W3 material or the maker's data. (No standard's table is reproduced.)
A7. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A8. λ (Thermal conductivity of the insulation) is taken as 0.035 W/(m*K). (Closed-cell elastomeric insulation; take the maker's value at the mean temperature (W3 material))
A9. $h_{mode}$ (Where the outer surface coefficient comes from) is taken as $h_{se}$ as given. (A surface coefficient is given, not computed.)
A10. $h_{se}$ (Outer surface coefficient $h_{se}$ (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)) is taken as 8 W/(m^2*K). (Still air in a plant room; replace with the project's own record.)
A11. eps (Emissivity of the outer surface (for the correlation; with $h_{se}$ as given it only judges that value against condensation on a cold surface)) is taken as 0.9. (Painted or plastic jacket (invented for this example); with $h_{se}$ given it only judges that value against condensation.)
A12. $v_{wind}$ (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Not used: $h_{se}$ is given.)
A13. $L_{surf}$ (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)

## 4. Working

[11] $\theta_{dew}$ = inverse Magnus of RH·$p_{sat}(\theta_a)$ [3]((Magnus form, as B7))
= $\theta_a$ = 28 °C, RH = 70 %
= 22.012 degC
Dew point of the ambient air

[12] $t_q$ = t such that |q| ≤ 12 W/m for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 17.814 mm
Required thickness, heat-gain limit

[13] $t_{dew}$ = t such that $\theta_{se}$ ≥ 23.012 °C for every thicker layer [1]
= scan 0 … 1000 mm, then bisection
= 13.162 mm
Required thickness, anti-condensation margin

[14] thickness = max over the selected criteria [1]
= $t_q$ = 17.814 mm, $t_{dew}$ = 13.162 mm
= 17.814 mm
Governing required thickness (heat-gain limit)

[16] $D_e = D_i + 2t$
= 90 mm + 2 × 25 mm
= 140 mm
Outer diameter of the insulation, chosen thickness

[17] $R_{ins} = \ln(D_e/D_i)/(2\pi\cdot\lambda)$ [1]
= ln(140/90)/(2π × 0.035)
= 2.0091 m·K/W
Resistance of the insulation

[18] $R_{se} = 1/(h_{se}\cdot\pi\cdot D_e)$ [1]
= 1/(8 × π × 0.14 m)
= 0.28421 m·K/W
Resistance of the outer surface

[19] $R = R_{oth} + R_{ins} + R_{se}$ [1]
= 0 + 2.0091 + 0.28421
= 2.2933 m·K/W
Total resistance per metre

[20] $q_{per,m} = |\theta_i - \theta_a| / R$ [1]
= 22 K / 2.2933
= 9.593 W/m
Heat flow per metre of pipe (heat gain, magnitude)

[21] $\theta_{se} = \theta_a + q/(h_{se}\cdot\pi\cdot D_e)$ [1]
= 28 °C − 9.593 / 3.5186
= 25.274 degC
Outer surface temperature

[22] $\theta_m = (\theta_i + \theta_{se})/2$
= (6 + 25.274)/2
= 15.637 degC
Mean temperature of the insulation: take λ at this temperature

[23] $D_{crit} = 2\lambda/h_{se}$ at the bare surface [1]
= 2 × 0.035 / 8
= 8.75 mm
Critical diameter at the bare-surface $h_{se}$: below it, insulation raises the heat flow

[24] $u_q = |q| / q_{max}$
= |q| ≤ 12 W/m
= 0.79941
Utilisation, heat-gain limit

[25] $u_{dew} = (\theta_a - \theta_{se})/(\theta_a - (\theta_{dew} + margin))$
= $\theta_{se}$ ≥ 23.012 °C
= 0.54661
Utilisation, anti-condensation margin

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\theta_{dew}$ | 22.012 degC | Dew point of the ambient air |
| $t_q$ | 17.814 mm | Required thickness, heat-gain limit |
| $u_q$ | 0.79941 | Utilisation, heat-gain limit |
| $t_{dew}$ | 13.162 mm | Required thickness, anti-condensation margin |
| $u_{dew}$ | 0.54661 | Utilisation, anti-condensation margin |
| thickness | 17.814 mm | Governing required thickness |
| $q_{per,m}$ | 9.593 W/m | Heat flow per metre (heat gain) |
| $\theta_{se}$ | 25.274 degC | Outer surface temperature |
| $\theta_m$ | 15.637 degC | Mean insulation temperature |
| R | 2.2933 m·K/W | Total resistance |
| $D_{crit}$ | 8.75 mm | Critical diameter (at the bare-surface $h_{se}$) |
| governing | "heat-gain limit" | Governing criterion |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Chosen thickness against the heat-gain limit | $u_q$ = 0.79941 | $u_q \le 1$ | 0.799 | [1] |
| PASS | Chosen thickness against the anti-condensation margin | $u_{dew}$ = 0.54661 | $u_{dew} \le 1$ | 0.547 | [1] |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: λ (Material properties); $h_{se}$ (Outer surface coefficient); eps (Surface emissivity); $q_{max,m}$, $dew_{margin}$ (Insulation criteria). Enter each as one so its source, edition and licence travel with it.

## 8. Verification

Template bs.insulation@1.1.1 Content hash 16a3b3c2a998a5e8 Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. ISO 12241.
2. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
3. ISO 13788.